

Introduction to stochastic processes: Markov Chains

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University of Washington - Seattle

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PROFESSIONAL & CONTINUING EDUCATION

UNIVERSITY *of* WASHINGTON



A Stochastic Processes Is:

- Any process in which outcomes in some variable (usually time, sometimes space, sometimes something else) are uncertain and best modelled probabilistically.
- **stochastic** is to **deterministic** as **random variable** is to **number**
- Stochastic processes can be continuous or discrete in time (index) and/or state.
 - “time series” (e.g. ARMA models) are usually discrete time-continuous state.
 - “Markov chains” (today’s topic) are usually discrete state.

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Examples include:

Just about everything.

Just about everything includes:

- Weather/Climate
- Population biology
 - Birth/death/reproduction/mortality
 - Migrations and movements
- Evolution
 - Population genetics (Mutation/Selection/Drift)
 - Gene sequences
- Epidemiology
 - Disease spread within a population (SIR models)
 - Disease spread within an organism
 - Development of resistance
- Tools for estimating models
 - MCMC (Markov Chain Monte Carlo)
 - Simulated annealing
- and much, much more

Just about anything ALSO includes:

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-
-

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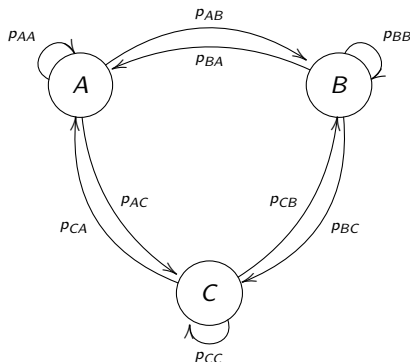
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All classic problems in stochastic processes!

Discrete state transitions

Consider $\mathbf{X} = \{X_1, X_2, X_3, \dots, X_n\}$ is in some discrete **state space** $\mathcal{E}$ (here: **A, B, C**) with fixed probabilities of transitioning from one state to another:



Sample sequence: $\mathbf{X} = \text{CCCBBCACCBABCBA...}$

This object is called a **Markov** chain.

Some definitions

$\mathbf{X}_n$ has the **Markov Property** if:

$$\Pr\{X_n = x_n | X_1 = x_1, \dots, X_{n-1} = x_{n-1}\} = \Pr\{X_n = x_n | X_{n-1} = x_{n-1}\}$$

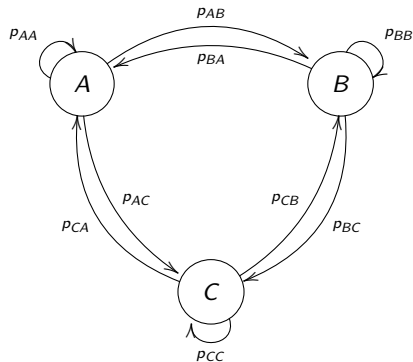
for all n in $x_1, \dots, x_n$.

In other words, any system whose future depends *only* on the present and not on the past has the *Markov Property* and any $\mathbf{X}_n$ that has the Markov property is a **Markov Chain**.

The $p_{ij}(t)$'s of a Markov chain are transition probabilities. If $p_{ij}(t)$'s are time invariant ($p_{ij}(t) = p_{ij}$) the chain is called **time homogeneous** or **stationary**.

Discrete state transitions

We express this process in terms of a **Probability Transition** matrix:


$$\mathbf{M} =$$

from: \ to:	A	B	C
A	p_{AA}	p_{AB}	p_{AC}
B	p_{BA}	p_{BB}	p_{BC}
C	p_{CA}	p_{CB}	p_{CC}

Such that:

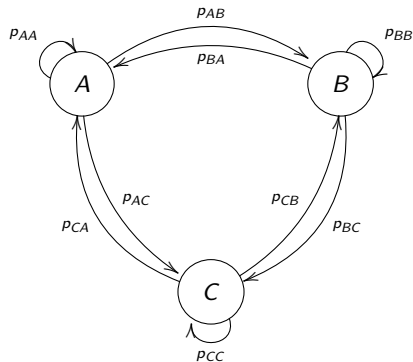
$$M_{ij} = \Pr(X_{t+1} = j | X_t = i) = p_{ij} \quad (1)$$

Note that:

$$\sum_{j=1}^n p_{ij} = 1 \dots \text{BUT} \dots \sum_{i=1}^n p_{ij} \neq 1 \quad (2)$$

Discrete state transitions

We express this process in terms of a **Probability Transition** matrix:



$$\mathbf{M} = \begin{array}{c|ccc} & \text{from:} \backslash \text{to:} & \text{A} & \text{B} & \text{C} \\ \hline \text{A} & & p_{AA} & p_{AB} & p_{AC} \\ \text{B} & & p_{BA} & p_{BB} & p_{BC} \\ \text{C} & & p_{CA} & p_{CB} & p_{CC} \end{array}$$

Such that:

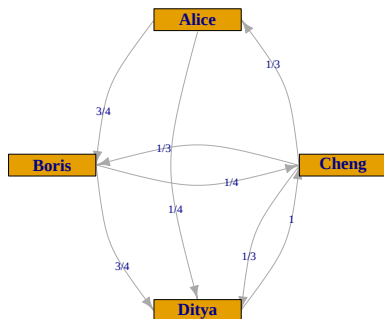
$$\Pr(X_{t+1} = j) = \sum_{i=1}^N M_{ij} \Pr(X_t = i) \quad (3)$$

Which can be conveniently rewritten in matrix notation as:

$$\pi_{t+1} = \mathbf{M} \times (\pi_t)^T \quad (4)$$

Where π_t is the distribution of the system over all states at time t .

Example 1: Children play frisbee



$$M = \begin{array}{c|cccc} & A & B & C & D \\ \hline A & 0 & 3/4 & 0 & 1/4 \\ B & 0 & 0 & 1/4 & 3/4 \\ C & 1/3 & 1/3 & 0 & 1/3 \\ D & 0 & 0 & 1 & 0 \end{array}$$

Let's give the ball to Alice, and see what happens:

A - B - D - C - D - C - B - D - C - D - C - B - C - D - C - D - C - B - C - A - B - D - C - B - C -
 B - D - C - B - D - C - A - D - C - D - C - D - C - A - D - C - A - B - C - D - C - A - B - D - C -
 - D - C - A - B - C - B - D - C - B - D - C - A - B - D - C - B - D - C - etc ...

This is called a **realization** of a stochastic process.

Simulating a Markov chain

```
M <- rbind(c(0,3/4,0,1/4), c(0,0,1/4,3/4), c(1/3,1/3,0,1/3), c(0,0,1,0))
rownames(M) <- c("Alice", "Boris", "Cheng", "Ditya")
colnames(M) <- rownames(M)
rowSums(M)

## Alice Boris Cheng Ditya
##      1      1      1      1

State<-1:4
X <- 1
Xs <- X
for(i in 1:30){
  X<-sample(State,1,prob=M[X,])
  Xs <- c(Xs, X)
}
paste(LETTERS[Xs], collapse="-")

## [1] "A-B-D-C-A-B-C-D-C-A-D-C-B-D-C-D-C-D-C-A-D-C-D-C-B-D-C-A-B-D-C"
```

Simulating a Markov chain

Let's make a subtle change

```
M[4,] <- c(0,0,0,1)
X <- 1
  Xs <- X
for(i in 1:30)
{
  X<-sample(State,1,prob=M[X,])
  Xs <- c(Xs, X)
}
paste(LETTERS[Xs], collapse="-")

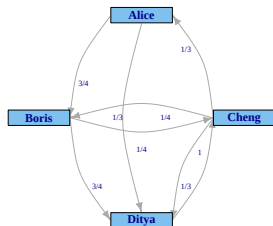
## [1] "A-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D-D"
```

What happened?

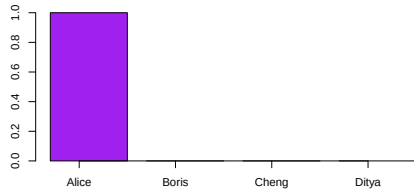
Consider the process probabilistically

Starting with Alice again:

$$\pi_0 = (1, 0, 0, 0)$$



1

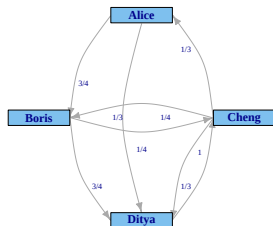


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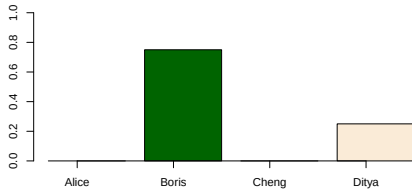
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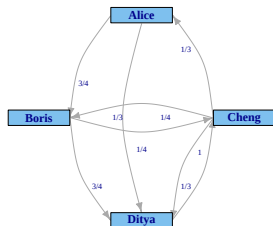
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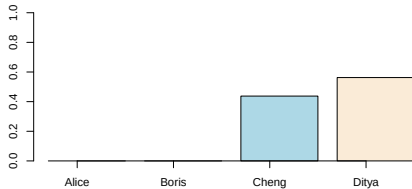
$$\pi_0 = (1, 0, 0, 0)$$

$$\pi_1 = (0, 0.75, 0, 0.25)$$

$$\pi_2 = (0, 0, 0.438, 0.562)$$



3



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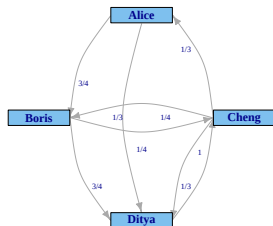
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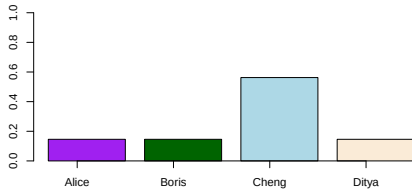
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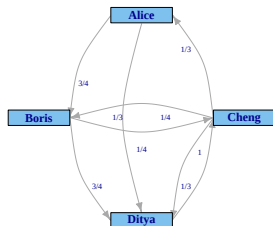
$$\pi_3 = (0.146, 0.146, 0.562, 0.146)$$



4



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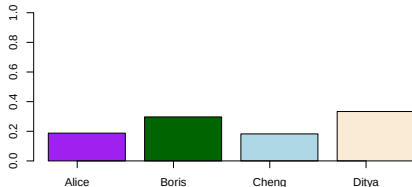
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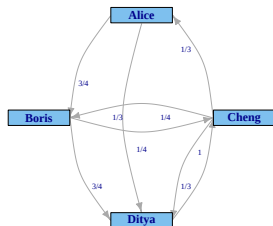
$$\pi_3 = (0.146, 0.146, 0.562, 0.146)$$

$$\pi_4 = (0.061, 0.201, 0.408, 0.33)$$

5



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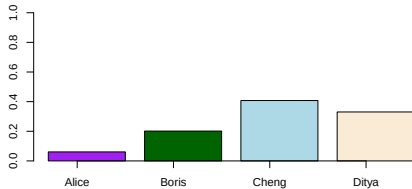
$$\pi_2 = (0, 0, 0.438, 0.562)$$

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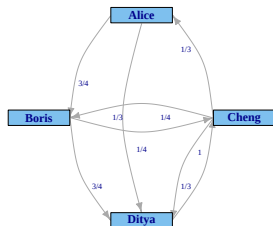
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$$\pi_5 = (0.136, 0.181, 0.381, 0.302)$$

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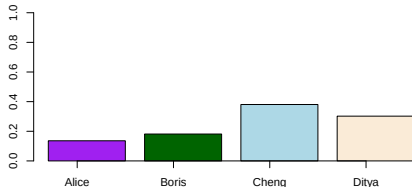
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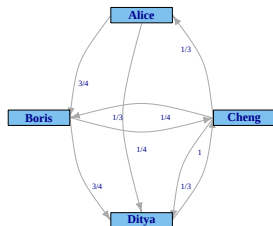
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$$\pi_6 = (0.127, 0.229, 0.347, 0.297)$$

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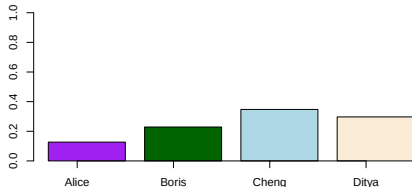
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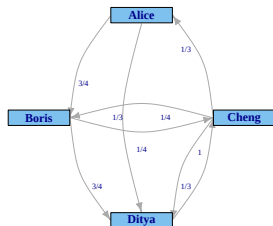
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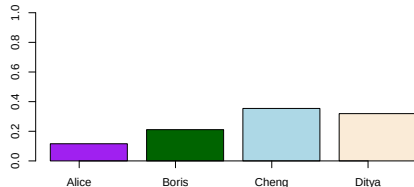
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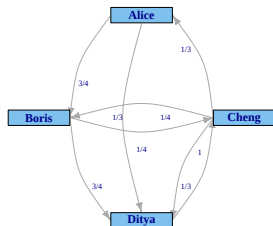
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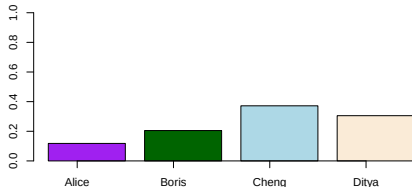
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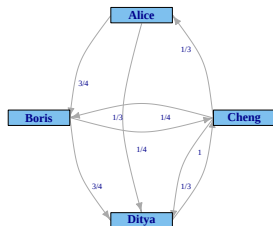
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10



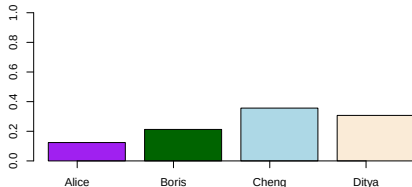
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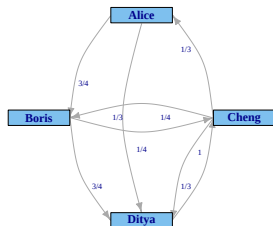
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11



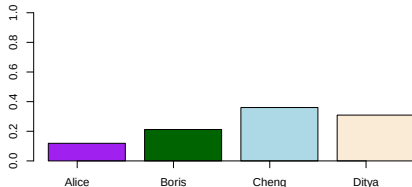
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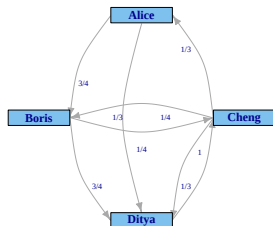
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12



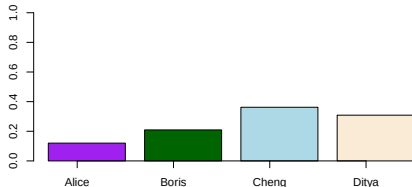
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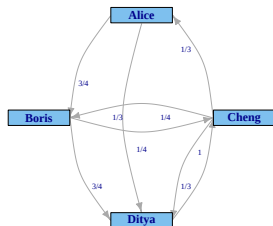
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 \end{aligned}$$

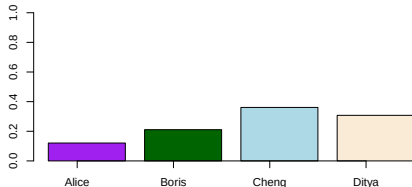
13



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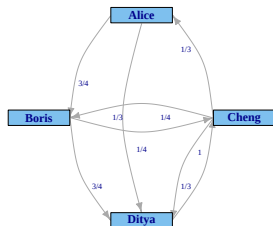
14



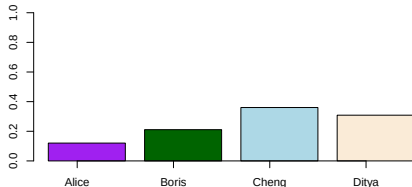
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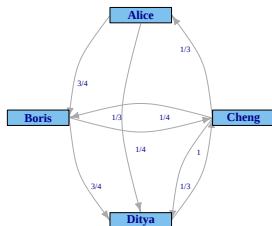


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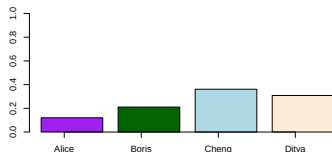
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 \pi_9 &= (0.124, 0.212, 0.356, 0.307) \\
 \pi_{10} &= (0.119, 0.212, 0.36, 0.309) \\
 \pi_{11} &= (0.12, 0.209, 0.362, 0.309) \\
 \pi_{12} &= (0.121, 0.211, 0.361, 0.308) \\
 \pi_{13} &= (0.12, 0.211, 0.36, 0.309) \\
 \pi_{14} &= (0.12, 0.21, 0.361, 0.308)
 \end{aligned}$$

The stationary state

The state: $\pi^* = (0.12, 0.21, 0.361, 0.308)$ is referred to as **stationary**. Note:



20



- 1 The name can be a bit confusing: the ball is not stationary, it is always moving around.
- 2 The state can be solved for mathematically:

$$\pi^* = \mathbf{M} \times (\pi^*)^T \quad (5)$$

This is a straightforward linear algebra problem, and is usually easy to obtain

- 3 All states have a value between 0 and 1 and have finite probability of being revisited forever and ever until the children's arms fall off. Such states are termed **recurrent**, **persistent** or **ergodic**.

Finding stationary state probabilities

Via matrix multiplication:

```
powermat=function(M,t){  
  Mt <- M  
  for(i in 2:t) Mt=Mt%%M  
  Mt}  
powermat(M,100)  
  
##           Alice      Boris      Cheng      Ditya  
## Alice 0.1203008 0.2105263 0.3609023 0.3082707  
## Boris 0.1203008 0.2105263 0.3609023 0.3082707  
## Cheng 0.1203008 0.2105263 0.3609023 0.3082707  
## Ditya 0.1203008 0.2105263 0.3609023 0.3082707
```

Via matrix algebra:

```
n <- nrow(M)  
a <- rbind((t(M) - diag(n))[-n,], rep(1,n))  
b <- c(rep(0,n-1),1)  
solve(a, b)  
  
##           Alice      Boris      Cheng      Ditya  
## 0.1203008 0.2105263 0.3609023 0.3082707
```

Why does this work?

$$\pi^* = M(\pi^*)^T$$

$$0 = (M - I)(\pi^*)^T$$

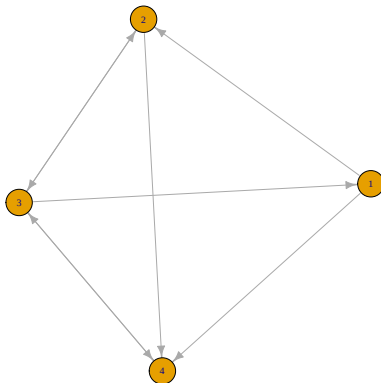
but the above is singular, so remove one element from the solution and add condition:

$$1 = \sum \pi^*$$

Note that `solve`, which we've used to invert matrices, also (or mainly) solves systems of equations expressed as matrices and vectors.

Graphing a Markov matrix

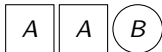
```
require(igraph)
M <- rbind(c(0,.75,0,.25), c(0,0,.25,.75), c(1/3,1/3,0,1/3), c(0,0,1,0))
M.graph <- graph.adjacency(M, weighted=TRUE)
plot(M.graph)
```



Check out the very cool `tkplot()` and `rglplot()` functions.

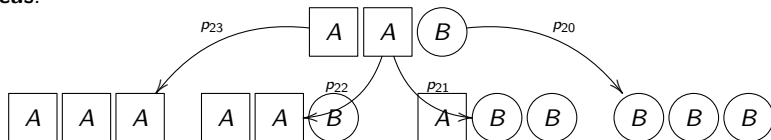
Example 2: Genetic Drift

Consider a population of size N that features 2 **alleles** (A and B) at a given **locus**.



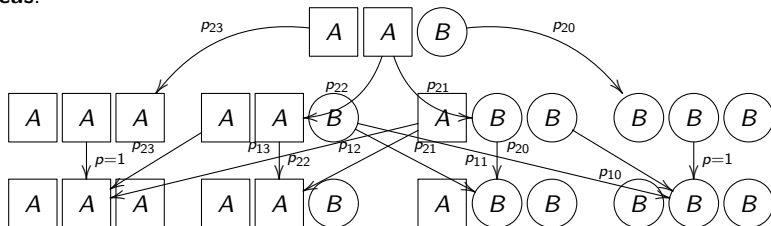
Example 2: Genetic Drift

Consider a population of size N that features 2 **alleles** (A and B) at a given **locus**.



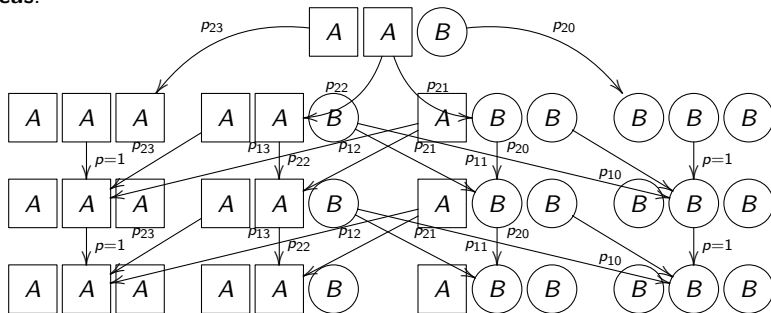
Example 2: Genetic Drift

Consider a population of size N that features 2 **alleles** (A and B) at a given **locus**.



Example 2: Genetic Drift

Consider a population of size N that features 2 **alleles** (A and B) at a given **locus**.



Genetic-Drift: Fisher-Wright Matrix

If the State X is defined as number of A alleles in the population, then:

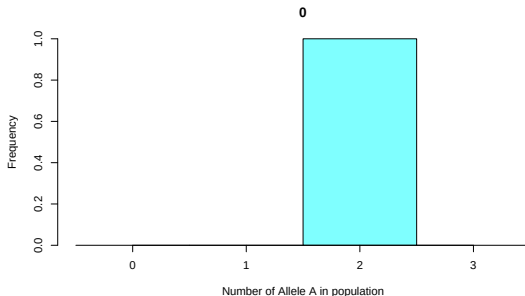
$$\mathbf{M} = \begin{array}{c} \begin{matrix} & 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} \left(\begin{array}{cccc} 1 & 0 & 0 & 0 \\ \left(\frac{2}{3}\right)^3 & 3\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^2 & 3\left(\frac{2}{3}\right)\left(\frac{1}{3}\right)^2 & \left(\frac{1}{3}\right)^3 \\ \left(\frac{1}{3}\right)^3 & 3\left(\frac{2}{3}\right)\left(\frac{1}{3}\right)^2 & 3\left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^2 & \left(\frac{2}{3}\right)^3 \\ 0 & 0 & 0 & 1 \end{array} \right) \end{array}$$

$$\mathbf{M} = \begin{array}{c} \begin{matrix} & 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} \left(\begin{array}{cccc} 1.000 & 0.000 & 0.000 & 0.000 \\ 0.296 & 0.444 & 0.222 & 0.037 \\ 0.037 & 0.222 & 0.444 & 0.296 \\ 0.000 & 0.000 & 0.000 & 1.000 \end{array} \right) \end{array}$$

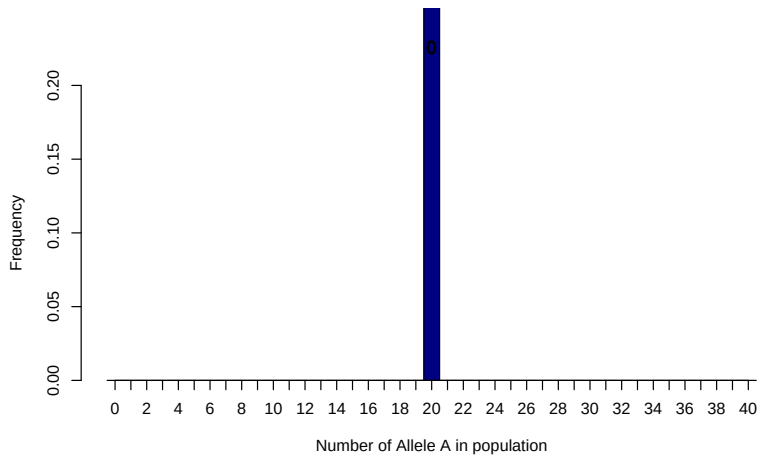
N=4

```
MakeFW<- function(N){  
  M <- matrix(NA,N+1,N+1)  
  for(i in 1:(N+1))  
    M[i,] <- dbinom(0:N,N,(i-1)/N)  
  return(M)  
}
```

```
N <- 4  
M <- MakeFW(N)  
X <- rep(0,N+1)  
X[round(N/2)+1] <- 1  
for(i in 1:20){  
  barplot(as.vector(X),main=i-1)  
  X<-X%*%M  
}
```



$N=40$



Fixation and transience

General Fisher-Wright matrix:

$$p_{ij} = Pr\{A_{t+1} = j | A_t = i\} = \binom{2N}{j} \left(\frac{i}{2N}\right)^j \left(1 - \frac{i}{2N}\right)^{2N-j} \quad (6)$$

The Fisher-Wright matrix is an **absorbing matrix**, and the states 0% and 100% are **absorbing states**. Other states are called **transient** (contrast with **recurrent**), because the process does not necessarily return to them.

Some properties of genetic drift:

- Always eventually fixates at 0 or N .
- Proportion of fixation depends on initial proportion of a given allele.
- Rate of fixation depends inversely on N

The final moral:

- **Genetic drift** is a Markovian fluctuation in allele frequencies that leads inexorably to fixation for small populations, but is counteracted by **mutation** and **migration** for large populations.

Historical aside on stochastic processes

Andrei Andreevich Markov (1856-1922) was a Russian mathematician who came up with the most widely used formalism and much of the theory for stochastic processes.

A passionate pedagogue, he was a strong proponent of problem-solving over seminar-style lectures. A politically principled activist, he refused tsarist honors, requested that he be excommunicated from the Russian Orthodox Church out of solidarity with the recently excommunicated **Leo Tolstoy**, publicly renounced his “membership in the electorate” when Parliament was dissolved, and eventually left his teaching post when the government demanded that teachers spy on students.



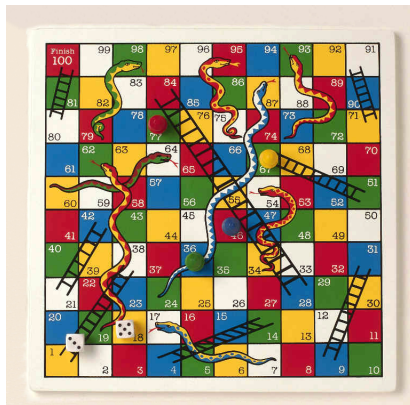
He said this of his most famous English colleague: *“I can judge all work only from a strictly mathematical point of view and from this viewpoint it is clear to me that ... Pearson has done nothing of any note.”*¹

¹from: Basharin et al. (2005) The Life and Work of A.A. Markov

Snakes and Ladders

An adorable example by Arthur Charpentier, from:

<http://www.r-bloggers.com/basics-on-markov-chain-for-parents/>



Snakes and Ladders

The Transition Matrix:

```
n=100

M=matrix(0,n+1,n+1+6)
rownames(M)=0:n
colnames(M)=0:(n+6)

for(i in 1:6)
{diag(M[, (i+1):(i+1+n)])=1/6}

M[,n+1]=apply(M[, (n+1):(n+1+6)],1,sum)

M=M[,1:(n+1)]

starting=c(4,9,17,20,28,40,51,54,62,
64,63,71,93,95,92)
ending  =c(14,31,7,38,84,59,67,34,19,
60,81,91,73,75,78)

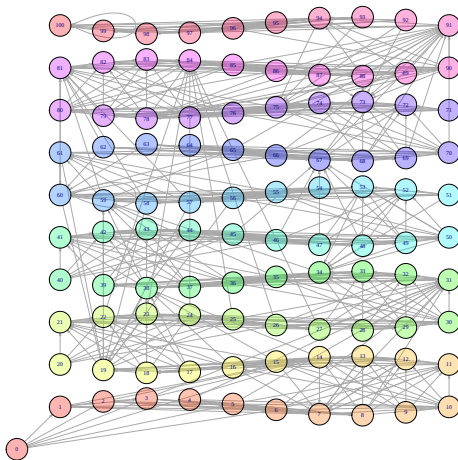
for(i in 1:length(starting)){
v=M[,starting[i]+1]
ind=which(v>0)
M[ind,starting[i]+1]=0
M[ind,ending[i]+1]=M[ind,ending[i]+1]+v[ind] }

row.names(M) <- 0:100
M.graph <- graph.adjacency(M, weighted=TRUE)
```

Snakes and Ladders

```
xs <- c(0,rep(c(1:10,10:1),5))
ys <- c(0,rep(1:10, each=10) + sin(seq(0,2*pi,length=10))/5)

plot(M.graph, layout = cbind(xs,ys), edge.arrow.size = 0.2, vertex.size = 10,
     vertex.color = rainbow(101, alpha=0.3), vertex.label.cex=0.5)
```



What can the Transition Matrix tell us?

The distribution after t turns:

$$X_t = X_0 \mathbf{M}^t$$

(where X_0 is $\{1, 0, 0, \dots\}$).

Position after 1 turns

100	99	98	97	96	95	94	93	92	91
81	82	83	84	85	86	87	88	89	90
80	79	78	77	76	75	74	73	72	71
61	62	63	64	65	66	67	68	69	70
60	59	58	57	56	55	54	53	52	51
41	42	43	44	45	46	47	48	49	50
40	39	38	37	36	35	34	33	32	31
21	22	23	24	25	26	27	28	29	30
20	19	18	17	16	15	14	13	12	11
1	2	3	4	5	6	7	8	9	10

What can the Transition Matrix tell us?

Your daughter loses her piece on the 10th move, and insists it was on square 60. Here are the conditional probabilities that it is one of 58, 59 or 60:

```
pi.0<- c(1,rep(0,100))  
(pi.0*%powermat(M,10))[59:61] %>% {./sum(.)}  
  
## [1] 0.1597003 0.5168209 0.3234788
```


How long you can expect to play?

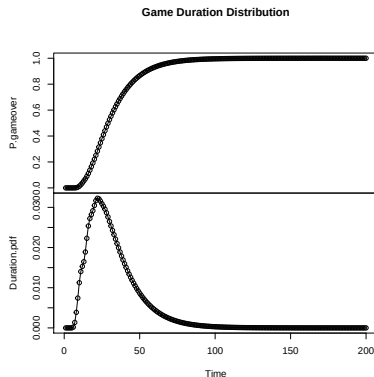
```
pi.t = pi.0%*%M

pi.t = pi.0 %*% M
P.gameover <- rep(NA,200)
for(move in 1:200){
  P.gameover[move] = pi.t[101]
  pi.t = pi.t %*% M
}

Duration.pdf <- diff(P.gameover)
sum(Duration.pdf * 1:length(Duration.pdf))

## [1] 32.16405
```

```
plot.ts(cbind(P.gameover,Duration.pdf),
        type="o", main="Game Duration Distribution")
```



These are “extinction” (or “gambler’s ruin”) problems.

How long will you be playing with two or three players?

```
sum((1-P.gameover )^2)
```

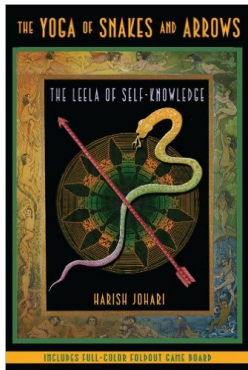
```
## [1] 23.40439
```

```
sum((1-P.gameover )^3)
```

```
## [1] 20.02098
```

With M, everything is Quick and Easy!

Snakes and Ladders: apparently much more than adorable!

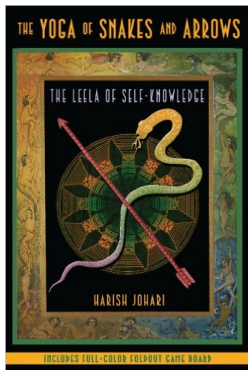


How long is a game of snakes and ladders?

S.C. ALTHOEN, L. KING and K. SCHILLING

“There is really only one Game, the Game in which each of us is a player acting out his role. The Game is Leela, the universal play of cosmic energy.” Thus begins Harish Johari in *Leela: the game of self-knowledge*, a serious commentary and religious interpretation of the Hindu board game *Leela*. *Leela* is essentially the game *Snakes and Ladders*, which in the U.S. is the popular children’s game *Chutes and Ladders*, first marketed as “the improved new version of snakes and ladders, England’s famous indoor sport,” for 50¢ in 1943 by the Milton Bradley Company of Springfield, Massachusetts. This game is really a 101 state absorbing Markov chain, which is amenable to mathematical as well as moral analysis. In what follows we investigate only the mathematical side of this diversion, specifically the expected playing time.

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